INCH-POUND

MIL-P-10388H
24 September 1990
SUPERSEDING
MIL-P-10388G
20 April 1982

MILITARY SPECIFICATION

PIPE FITTINGS: ONE OR MORE ENDS GROOVED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers grooved- and combination-type pipe fittings.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640	- Box, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60	- Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730

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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-10387	- Couplings, Clamp, Pipe: With Bolts and Synthetic Rubber Gaskets for Grooved-End Pipe.
MIL-P-11087	- Pipe, Steel: Grooved, Threaded, or Plain Ends.
MIL-C-46168	- Coating, Aliphatic, Polyurethane, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1	- Pipe Threads, General Purpose (Inch).
ASME B16.1	- Cast-Iron Pipe Flanges and Flanged Fittings (Class 25, 125, and 800).
ASME B16.5	- Pipe Flanges and Flanged Fittings.
ASME B16.25	- Buttwelding Ends.
ASME B46.1	- Surface Texture.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A47	- Ferritic Malleable Iron Castings.
ASTM A536	- Ductile Iron Castings.
ASTM B26	- Aluminum Alloy Sand Castings.
ASTM B210	- Aluminum and Aluminum-Alloy Drawn Seamless Tubes.
ASTM B247	- Aluminum-Alloy Die Forgings, Hand Forgings and Rolled Ring Forgings.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Fittings shall be malleable iron, ductile iron, steel, or aluminum alloy, and shall be furnished with various combinations of flanged, threaded, beveled, and grooved ends. All fittings shall have one or more ends grooved.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all the provisions of this specification.

3.3.1 Material deterioration prevention and control. The pipe fittings shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the pipe fittings may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pipe fittings may be newly fabricated from recovered materials to the maximum extent practicable, provided the pipe fittings produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the fabrication of pipe fittings.

3.3.3 Adapters and flanged reducer fittings.

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3.3.3.1 Aluminum flanged fittings. Aluminum flanged fittings shall be fabricated from round aluminum-alloy tubing conforming to ASTM B 210, alloy UNS A96061, T6, wall thickness to fit the application, and flanges of aluminum-alloy forgings conforming to ASTM B247 alloy UNS A96151, or aluminum-alloy castings conforming to ASTM B26, alloy and temper to fit the application.

3.3.3.2 Steel flanged fittings. Steel flanged fittings shall be fabricated from steel pipe conforming to MIL-P-11087, size and weight as shown in table IX, and flanges of steel selected by the contractor subject to all the provisions of this specification.

3.3.3.3 Iron flanged fittings. Iron flanged fittings shall be fabricated from steel pipe conforming to MIL-P-11087, size and weight as shown in table IX, and flanges of malleable iron castings conforming to ASTM A 47, grade 32510, or ductile iron castings conforming to ASTM A 536, grade 65-45-12.

3.3.4 Couplings.

3.3.4.1 Steel couplings. Steel couplings shall be steel pipe conforming to MIL-P-11087, size and weight as shown in table I.

TABLE I. Diameter and weight.

Nominal pipe size (inches)	Outside pipe diameter (inches)	Minimum weight (lb/ft)
3/4	1.050	1.13
1	1.315	1.68
1-1/4	1.660	2.27
1-1/2	1.900	2.72
2	2.375	3.65
2-1/2	2.875	5.79
3	3.500	7.58
3-1/2	4.000	9.11
4	4.500	10.79
5	5.563	14.62
6	6.625	18.97
8	8.625	24.70
10	10.750	28.04
12	12.750	33.38
14	14.000	36.71

3.3.4.2 Aluminum couplings. Aluminum couplings shall be round aluminum-alloy tube conforming to ASTM B 210, alloy UNS A96061, T6, wall thickness to fit the application.

3.3.5 Fittings (except adapter, coupling, and flanged reducer).

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3.3.5.1 Iron fittings. Fittings shall be malleable iron castings conforming to ASTM A 47, grade 32510, or ductile iron castings conforming to ASTM A 536, grade 65-45-12.

3.3.5.2 Steel fittings. Steel fittings shall be steel selected by the contractor and subject to all provisions of this specification.

3.3.5.3 Aluminum fittings. Aluminum fittings shall be aluminum-alloy forgings conforming to ASTM B 247, alloy UNS A96151, or aluminum-alloy castings conforming to ASTM B 26, alloy and temper to fit the application.

3.3.6 Fitting ends.

3.3.6.1 Beveled. Beveled ends shall conform to ASME B16.25.

3.3.7 Flanged ends.

3.3.7.1 Cast iron. Cast iron flanged ends shall conform to ASME B16.1, class 125 or 250.

3.3.7.2 Steel. Steel flanged ends shall conform to ASME B16.5, class 150 or 300.

3.3.7.3 Aluminum. Aluminum-alloy flanged ends shall be faced and drilled and conform to ASME B16.1, class 125, 250, or ASME B16.5, class 150 or 300.

3.3.8 Grooved ends. Grooved ends shall conform to the dimensions shown on figure 1. The land-sealing surface shall be smooth and free from mill marks and shall have a roughness height rating of not more than 500 and conform to ASME B46.1.

3.3.9 Threaded ends. Threaded ends shall be National Pipe Thread (NPT) and conform to ASME B1.20.1.

3.4 Design. Fittings shall be malleable iron, ductile iron, steel, or aluminum alloy, as specified (see 6.2 and 6.5.1). All fittings shall have one end grooved and the other end beveled, flanged, grooved, or threaded. The groove may be continuous or backstop lug design. As specified (see 6.2 and 6.5.1), fittings shall be drilled, tapped, and a pipe plug inserted for 0.25-inch, 0.50-inch, 0.75-inch, or 2-inch NPT connection.

3.4.1 Adapters and flanged reducers. Cast-iron adapters and flanged reducers shall consist of steel pipe threaded into a cast-iron flange. Steel adapters and flanged reducers shall consist of steel pipe threaded or welded to a steel flange. When pipe and flange are threaded together, the threads shall be coated with a nonhardening, nongalling compound inert to aromatic gasoline. Aluminum adapters and flanged reducers shall consist of aluminum-alloy tubing welded to an aluminum-alloy flange.

3.4.2 Tees. When specified (see 6.5.5) tees shall be furnished with a 1/2-inch cleaner guide. Material of cleaner guide for aluminum fittings shall be 1/2 inch aluminum. Material of cleaner guide for steel or ductile iron fittings shall be 1/2 inch steel. Cleaner guide shall be attached to the fitting by brazing or welding across the inside diameter of the branch outlet,

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with the inner side in line with the bore of the main outlets. Joining processes shall not reduce the strength of the fitting to the extent that the fitting could not withstand the desired hydrostatic test pressure. Appropriate filler material and flux shall be used to prevent corrosion.

3.5 Working and test pressures.

3.5.1 Adapter and flanged reducers. Adapter and flanged reducers shall withstand the applicable working and hydrostatic pressures shown in table II without evidence of leakage, rupture, or yielding.

TABLE II. Working and hydrostatic test pressures.

Class	Maximum working pressure (WOG)1/ (psi)	Hydrostatic test pressure (psi)
125	175	250
150	230	500
250	500	900
300	720	1100

1/ Water, oil, or gas

3.5.2 Fittings (except adapter and flanged reducers). All fittings except adapter and flanged reducers shall have a working pressure of not less than 600 pounds per square inch (psi) for water, oil, or gas and shall withstand the hydrostatic pressures specified in table III without evidence of leakage, rupture, or yielding.

TABLE III. Hydrostatic test pressures.

Nominal pipe size (inches)	Hydrostatic test pressure (psi)
3/4	2000
1	2000
1-1/4	2000
1-1/2	2000
2	2000
2-1/2	1800
3	1800
3-1/2	1800
4	1600
5	1600
6	1400
8	1200
10	1000
12	1000
14	800

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3.6 Dimensions.

3.6.1 Fitting lengths. Fitting dimensions shall conform to table IV for the size specified. Lengths are not given in table IV for adapters, couplings, and reducers. Therefore, adapter, coupling and reducer lengths must be specified as part of the specification part number (see 6.2 and 6.5.6).

TABLE IV. Fitting length.

<table>
<tr><th colspan="5">Adapters - one end flanged.</th></tr>
<tr><th rowspan="2">Nominal size (inches)</th><th colspan="4">Length of adapter (inches) measured end to end</th></tr>
<tr><th>Class 125</th><th>Class 150</th><th>Class 250</th><th>Class 300</th></tr>
<tr><td>3/4
1
1-1/4
1-1/2
2
2-1/2
3
3-1/2
4
5
6
8
10
12
14</td><td colspan="4">Length for all sizes and designations of flanged adapters shall be as specified (see 6.5.6).</td></tr>
<tr><td colspan="5">Cap</td></tr>
<tr><td colspan="5">Nominal size (inches) 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, 4, 5, 6, 8, 10, 12, and 14.</td></tr>
<tr><td colspan="5">Length: Seats flush with outside of coupling.</td></tr>
</table>

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TABLE IV. Fitting length - Continued.

Couplings				
	Length of coupling (inches) measured end to end			
Nominal size (inches)	Beveled one end	Threaded one end		Grooved both ends
		External	Internal	
3/4 1 1-1/4 1-1/2 2 2-1/2 3 3-1/2 4 5 6 8 10 12 14	Length for all sizes and designations of coupling shall be as specified (see 6.5.6).			

Cross	
Nominal size (inches)	Length of cross (inches) measured center to ends, all ends grooved
3/4	2-1/4
1	2-1/4
1-1/4	2-3/4
1-1/2	2-3/4
2	3-1/4
2-1/2	3-3/4
3	4-1/4
3-1/2	4-1/2
4	5
5	5-1/2
6	6-1/2
8	7-3/4
10	9
12	10
14	11

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TABLE IV. Fitting length - Continued.

Elbows									
Nominal size (inches)	Length of elbows (inches) Both ends grooved					External threaded one end			
	Center to end				End to end	Center to threaded end		Center to grooved end	
	90°	45°	22-1/2°	11-1/4°	Adjustable	90°	45°	90°	45°
3/4	2-1/4	1-1/2	1-5/8	1-3/8	-	2-1/4	1-1/2	2-1/4	1-1/2
1	2-1/4	1-3/4	1-5/8	1-3/8	3-1/4	2-1/4	1-3/4	2-1/4	1-3/4
1-1/4	2-3/4	1-3/4	1-3/4	1-3/8	-	2-3/4	1-3/4	2-3/4	1-3/4
1-1/2	2-3/4	1-3/4	1-3/4	1-3/8	-	2-3/4	1-3/4	2-3/4	1-3/4
2	3-1/4	2	1-7/8	1-3/8	3-3/4	4-1/4	3	3-1/4	2
2-1/2	3-3/4	2-1/4	2	1-1/2	4	3-3/4	2-1/4	3-3/4	2-1/4
3	4-1/4	2-1/2	2-1/4	1-1/2	4-1/2	6	4-1/4	4-1/4	2-1/2
3-1/2	4-1/2	2-3/4	2-1/2	1-3/4	5	6-1/4	2-3/4	4-1/2	2-3/4
4	5	3	2-5/8	1-3/4	5-1/4	7-1/4	5-1/4	5	3
5	5-1/2	3-1/4	2-7/8	2	-	-	-	-	-
6	6-1/2	3-1/2	3-1/8	2	6-1/4	6-1/2	3-1/2	6-1/2	3-1/2
8	7-3/4	4-1/4	3-7/8	2	7-3/4	-	-	-	-
10	9	4-3/4	4-3/8	2-1/8	-	-	-	-	-
12	10	5-1/4	4-7/8	2-1/4	-	-	-	-	-
14	21	8-3/4	5	3-1/2	-	-	-	-	-

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TABLE IV. Fitting length - Continued.

Nominal size (inches)	45° Lateral	
	Length of lateral (inches) All ends grooved	
	Center to long end	Center to short end
3/4	4-1/2	2
1	5	2-1/4
1-1/4	5-3/4	2-1/2
1-1/2	6-1/4	2-3/4
2	7	2-3/4
2-1/2	7-3/4	3
3	8-1/2	3-1/4
3-1/2	10	3-1/2
4	10-1/2	3-3/4
5	12-1/2	4
6	14	4-1/2
8	18	6
10	20-1/2	6-1/2
12	23	7
14	26-1/2	7-1/2

Reducers				
Nominal size (inches)	Length of reducers (inches) measured end to end			
	Small end threaded	Small end flanged	Large end flanged	Both ends grooved
1-1/2 x 1 2 x 3/4 2 x 1 2 x 1-1/4 2 x 1-1/2 2-1/2 x 2 3 x 1 3 x 2	Length for all sizes and designations of reducers shall be as specified (see 6.5.6).			

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TABLE IV. Fitting length - Continued.

Nominal size (inches)	Length of reducers (inches) measured end to end			
	Small end threaded	Small end flanged	Large end flanged	Both ends grooved
3 x 2-1/2 3-1/2 x 3 4 x 1 4 x 2 4 x 2-1/2 4 x 3 4 x 3-1/2 5 x 4 6 x 1 6 x 2 6 x 3 6 x 4 6 x 5 8 x 3 8 x 4 8 x 5 8 x 6 10 x 4 10 x 6 10 x 8 12 x 6 12 x 8 12 x 10				

Tees

Nominal size (inches)	Length of tees (inches) measured center to ends	
	Grooved all ends Reducing	
3-4 x 3/4 x 3/4 1 x 1 x 1 1-1/4 x 1-1/4 x 1-1/4 1-1/2 x 1-1/2 x 1-1/2 2 x 2 x 3/4 x 1 x 1-1/2 x 2	2-1/4 2-1/4 2-3/4 2-3/4 - - - 3-1/4	- - - - 3-1/4 3-1/4 3-1/4 -

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TABLE IV. Fitting length - Continued.

Tees		
Nominal size (inches)	Length of tees (inches) measured center to ends	
	Grooved all ends	Reducing
2-1/2 x 2-1/2 x 2-1/2	3-3/4	-
3 x 3 x 1	-	4-1/4
x 1-1/2	-	4-1/4
x 2	-	4/1/4
x 2-1/2	-	4-1/4
x 3	4-1/4	-
3-1/2 x 3-1/2 x 3-1/2	4-1/2	-
4 x 4 x 1	-	3-3/4
x 2	-	5
x 2-1/2	-	5
x 3	-	5
x 4	5	-
5 x 5 x 2	-	5-1/2
x 3	-	5-1/2
x 4	-	5-1/2
x 5	5-1/2	-
6 x 6 x 2	-	6-1/2
x 2-1/2	-	6-1/2
x 3	-	6-1/2
x 4	-	6-1/2
x 5	-	6-1/2
x 6	6-1/2	-
8 x 8 x 2	-	7-3/4
x 3	-	7-3/4
x 4	-	7-3/4
x 5	-	7-3/4
x 6	-	7-3/4
x 8	7-3/4	-
10 x 10 x 2	-	9
x 3	-	9
x 4	-	9
x 5	-	9
x 6	-	9
x 8	-	9
x 10	9	-
12 x 12 x 3	-	10
x 4	-	10
x 5	-	10

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TABLE IV. Fitting length - Continued.

Tees		
Nominal size (inches)	Length of tees (inches) measured center to ends	
	Grooved all ends	Reducing
14 x 14 x 6	-	10
14 x 14 x 8	-	10
14 x 14 x 10	-	10
14 x 14 x 12	10	-
14 x 14 x 8	-	11
14 x 14 x 10	-	11
14 x 14 x 12	-	11
14 x 14 x 14	11	-

Union (flexible)	
Nominal size (inches)	Length of union (inches) measured end to end both ends grooved
3/4	4
1	4-1/8
1-1/4	4-5/8
1-1/2	4-5/8
2	5
3	5-1/2
4	6-1/2

3.7 Treatment and painting. The portions of the fittings normally painted shall be cleaned, treated, and painted conforming to MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 and conform to MIL-C-46168.

3.8 Identification marking. Fittings shall be identified conforming to MIL-STD-130.

3.9 Workmanship. The fittings shall be free from pits, cracks, burrs, and any other defect that could impair serviceability.

3.9.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Aluminum welding shall be done by the inert gas (argon or helium), shielded arc method. Welds may be ground, filed, or buffed, but shall not be hammered.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall conform to MIL-STD-105.

4.4.2 Examination. Samples selected as specified in 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause of rejection.

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4.4.3 Tests. Samples selected as specified in 4.4.1 shall be tested conforming to 4.5.2. Any fitting failing to pass the test shall be considered defective.

4.5 Examination procedure.

4.5.1 Examination. The fittings shall be examined as specified herein for the following defects:

Major

101. Material not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration (see 3.3.1).
103. Dissimilar metals do not conform to MIL-STD-889 (see 3.3.1.1).
104. Material and material finish or treatment information not available (see 3.3.1.2).
105. Used, rebuilt or remanufactured components, pieces or parts incorporated in the pipe fittings (see 3.3.2).
106. Adapters and flanged reducer fittings not as specified (see 3.3.3).
107. Couplings not as specified (see 3.3.4).
108. Iron, steel and aluminum fittings not as specified (see 3.3.5).
109. Fitting ends not as specified (see 3.3.6).
110. Flanged ends not as specified (3.3.7).
111. Grooved ends not as specified (see 3.3.8).
112. Threaded ends not as specified (see 3.3.9).
113. Design not as specified (see 3.4).
114. Fitting lengths not as specified (see 3.6).
115. Treatment and painting not as specified (see 3.7).
116. Workmanship not as specified (see 3.9).
117. Welding not as specified (see 3.9.1).

Minor

201. Pipe plugs missing (see 3.4).
202. Cleaner guides misaligned in tees (see 3.4.2).
203. Identification marking not as specified (see 3.8).

4.5.2 Tests.

4.5.2.1 Hydrostatic pressure. The fitting shall be prepared for hydrostatic test by closing all openings with blank ends. One of the blank ends shall be equipped with a nipple or adapter through which recirculated fresh water containing a rust preventive can be pumped. The fitting shall be subjected to the applicable pressure as shown on tables X and XI for not less than 10 minutes. During this time, the fitting shall be examined. Any evidence of leakage, yielding, or rupture shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

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4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall conform to MIL-STD-105.

4.6.1.3 Examination. Samples as specified in 4.6.1.2 shall be examined for the following major defects. Presence of one or more defects shall be cause for rejection.

118. Land-sealing surface of grooved end fittings not completely covered with barrier material and tape as specified.
119. Application of barrier material and tape not in accordance with figure 2 as specified.
120. Fittings of unlike size or description packed together.
121. Boxes not as specified.
122. Strapping not as specified.
123. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be level A as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Land-seals shall be cleaned by the appropriate process in accordance with MIL-P-116.

5.1.1.2 Drying. Land-seals shall be dried by the appropriate process in accordance with MIL-P-116.

5.1.1.3 Preservation application. Preservatives shall not be used. The land-sealing surface of groove-end fittings shall be completely covered with a single wrap of barrier material conforming to MIL-B-121, type 1, class 2 (see figure 1). The wrap for the land-sealing surface shall be flush with the end of the fitting and shall extend back beyond the groove not less than 1/4 inch. The wrap of the barrier material shall be covered with a single wrap of tape conforming to PPP-T-60, type IV. The tape shall be a width that will completely cover the barrier material and shall extend not less than 1/8 inch beyond the end of the fitting and extend back beyond the barrier material not less than 1/2 inch. The end of the tape extending beyond the end of the fitting and the tape extending beyond the wrap shall be rolled in place against the end wall of the fitting and against the fitting (see figure 1).

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Fittings of like size and description shall be packed together in close-fitting boxes conforming to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.2.2 Level B. Fittings of like size and description shall be packed together in close-fitting boxes conforming to PPP-B-621, class 2, style optional; PPP-B-601, domestic type, style optional; or PPP-B-640, class 2,

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style optional. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable, finish A.

5.2.3 Level C. The fittings of like size and description shall be packed in containers in quantities not exceeding 200 pounds gross weight. Containers shall be closed and sealed with adhesive, reinforced tape, or strapping.

5.3 Marking. Marking shall conform to MIL-STD-129.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Grooved-type pipe fittings are intended for use as components of portable pipelines conveying light petroleum products or water. Couplings for grooved-type pipe are described in MIL-C-10387.

6.2 Ordering data. Acquisition documents should specify the following:

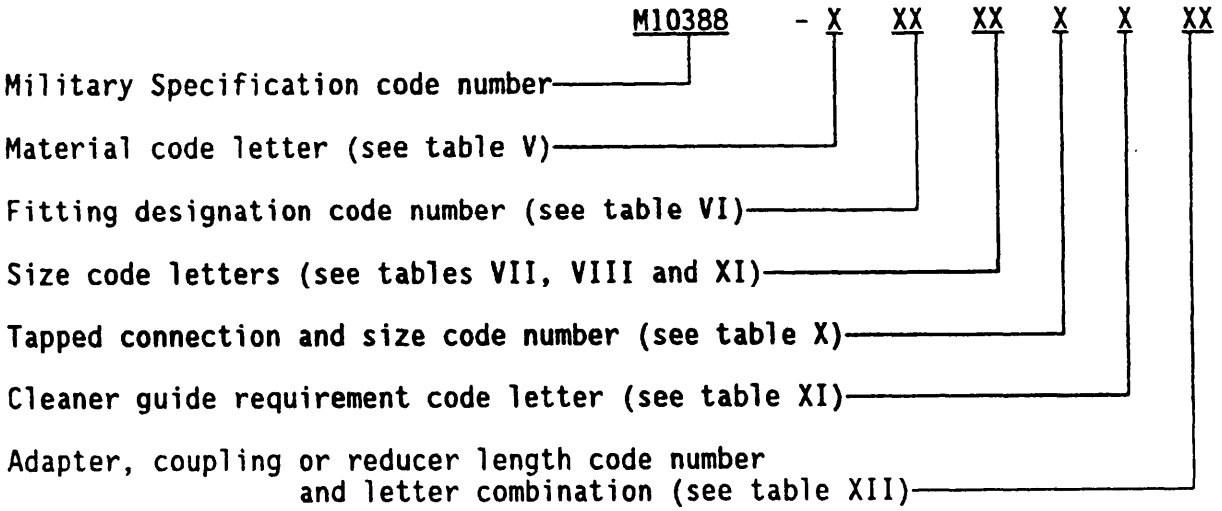
a. Title, number, and date of this specification.
b. Military specification part number required (see 6.5, 3.3, 3.4, 3.4.2, and 3.6.1).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is not required (see 3.2).
e. Color required (see 3.7).
f. Levels of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 International standardization agreement. The dimensions and tolerances shown in figure 2 are the subject of international standardization agreement (NATO STANAG 2761 and QSTAG-240). When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

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6.5 Specification part number. The part number of pipe fittings covered by this specification shall be designated in the following form (see 6.2).



6.5.1 Material. The material of the fitting is identified by a single letter (see table V).

TABLE V. Type material code letter.

Code Letter	Material
A	Aluminum alloy.
B	Malleable iron, ductile iron or steel.

6.5.2 Fitting designation. The kind of fitting is identified by a two-digit number (see table VI).

TABLE VI. Fitting designation code number.

Fitting kind	Fitting code number
ADAPTERS	
Flanged (class 125) one end.	01
Flanged (class 150) one end.	02
Flanged (class 250) one end.	03
Flanged (class 300) one end.	04
LATERALS	
45 degree all ends grooved.	18
REDUCERS	
Grooved both ends.	19
Flanged (class 125) large end.	20

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TABLE VI. Fitting designation code number - Continued.

Fitting kind	Fitting code number
CAP	05
COUPLINGS	
Beveled one end.	06
External threaded one end.	07
Grooved both ends.	08
Internal threaded one end.	09
CROSS	10
ELBOWS	
90 degree both ends grooved.	11
90 degree external threaded one end.	12
45 degree both ends grooved.	13
45 degree external threaded one end.	14
22-1/2 degree both ends grooved.	15
11-1/4 degree both ends grooved.	16
Adjustable both ends grooved.	17
Flanged (Class 150) large end.	21
Flanged (class 250) large end.	22
Flanged (class 300) large end.	23
Flanged (class 125) small end.	24
Flanged (class 150) small end.	25
Flanged (class 250) small end.	26
Flanged (class 300) small end.	27
Threaded small end.	28
TEES	
Grooved all ends.	29
Reduced grooved all ends.	30
Reducing run ends grooved, branch end external threaded.	32
Reducing run ends grooved, branch end internal threaded.	32
UNION flexible both ends grooved.	3

6.5.3 Size. The nominal pipe size of the fitting is identified by two letters. Sizes for all fittings (except reducers and tees) are covered in table VII. Reducer sizes are covered in table IV. Tee sizes are covered in table IX.

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TABLE VII. Size code letters (except reducers and tees).

Nominal pipe size (inches)	Size code letters
3/4	AA
1	AB
1-1/4	AC
1-1/2	AD
2	AE
2-1/2	AF
3	AG
3-1/2	AH
4	AI
5	AJ
6	AK
8	AL
10	AM
12	AN
14	AO

TABLE VIII. Reducer size code letters.

Nominal pipe size (inches)	Size code letters
1-1/2 x 1	BA
2 x 3/4	BB
x 1	BC
x 1-1/4	BD
x 1-1/2	BE
2-1/2 x 2	BF
x 2-1/2	BM
x 3	BN
x 3-1/2	BO
5 x 4	BP
6 x 1	BQ
x 2	BR
x 3	BS
x 4	BT
x 5	BU

Nominal pipe size (inches)	Size code letters
3 x 1	AG
x 2	BH
x 2-1/2	BI
3-1/2 x 3	BJ
4 x 1	BK
x 2	BL
8 x 3	BV
x 4	BW
x 5	BX
x 6	BY
10 x 4	BZ
x 6	CA
x 8	CB
12 x 6	CC
x 8	CD
x 10	CE

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TABLE IX. Tee size code letters.

Nominal pipe size (inches)	Size code letters
3/4 x 3/4 x 3/4	DA
1 x 1 x 1	DB
1-1/4 x 1-1/4 x 1-1/4	DC
1-1/2 x 1-1/2 x 1-1/2	DD
2 x 2 x 3/4	DE
x 1	DF
x 1-1/2	DG
x 2	DH
2-1/2 x 2-1/2 x 2-1/2	DI
3 x 3 x 1	DJ
x 1-1/2	DK
x 2	DL
x 2-1/2	DM
x 3	DN
x 4	EB
x 5	EC
x 6	ED
8 x 8 x 2	EE
x 3	EF
x 4	EG
x 5	EH
x 6	EI
x 8	EJ
10 x 10 x 2	EK
x 3	EL
x 4	EM
x 5	EN
x 6	EO

Nominal pipe size (inches)	Size code letters
3-1/2 x 3-1/2 x 3-1/2	DO
4 x 4 x 1	DP
x 2	DQ
x 2-1/2	DR
4 x 4 x 3	DS
x 4	DT
5 x 5 x 2	DU
x 3	DV
x 4	DW
x 5	DX
6 x 6 x 2	DY
x 2-1/2	DZ
x 3	EA
x 8	EP
x 10	DQ
12 x 12 x 3	ER
x 4	ES
x 5	ET
x 6	EU
x 8	EV
x 10	EW
x 12	EX
14 x 14 x 8	EY
x 10	EZ
x 12	FA
x 14	FB

6.5.4 Tapped connection and size. Fittings with a tapped connection and the size of the drilled and tap connection are identified by a single number (see table X).

TABLE X. Tapped connection and size code number.

Code number	Size of drilled and tapped connection (inches)
1	None
2	1/4
3	1/2
4	3/4
5	1
6	2

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6.5.5 Cleaner guide. A single code letter is used to designate a cleaner guide that is furnished with a tee (see table XI).

TABLE XI. Cleaner guide code letter.

Code letter	Cleaner guide requirement
A	Without cleaner guide (for tees).
B	With cleaner guide (for tees).
C	Without cleaner guide (for fittings other than tee).

6.5.6 Adapter, coupling and reducer length. The length of the adapter, coupling, or reducer shall be identified by a number and letter combination. The number shall be the number of whole inches in the required length, and the letter code shall indicate the fractional part of an inch of the required length (example: a length of 8.0 inches in 8A, 5.50 inches in 5C, 0.75 inch is 0D, and 10.25 inches is 10B) (see table XII).

TABLE XII. Fractional inch code letter.

Code Letter	Fractional Inch
A	0
B	1/4
C	1/2
D	3/4

6.6 Subject term (key word) listing.

Pipe fittings, aluminum alloy
Pipe fittings, beveled ends
Pipe fittings, ductile iron
Pipe fittings, flanged ends
Pipe fittings, grooved ends
Pipe fittings, malleable iron
Pipe fittings, steel

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - ME
Navy - YD

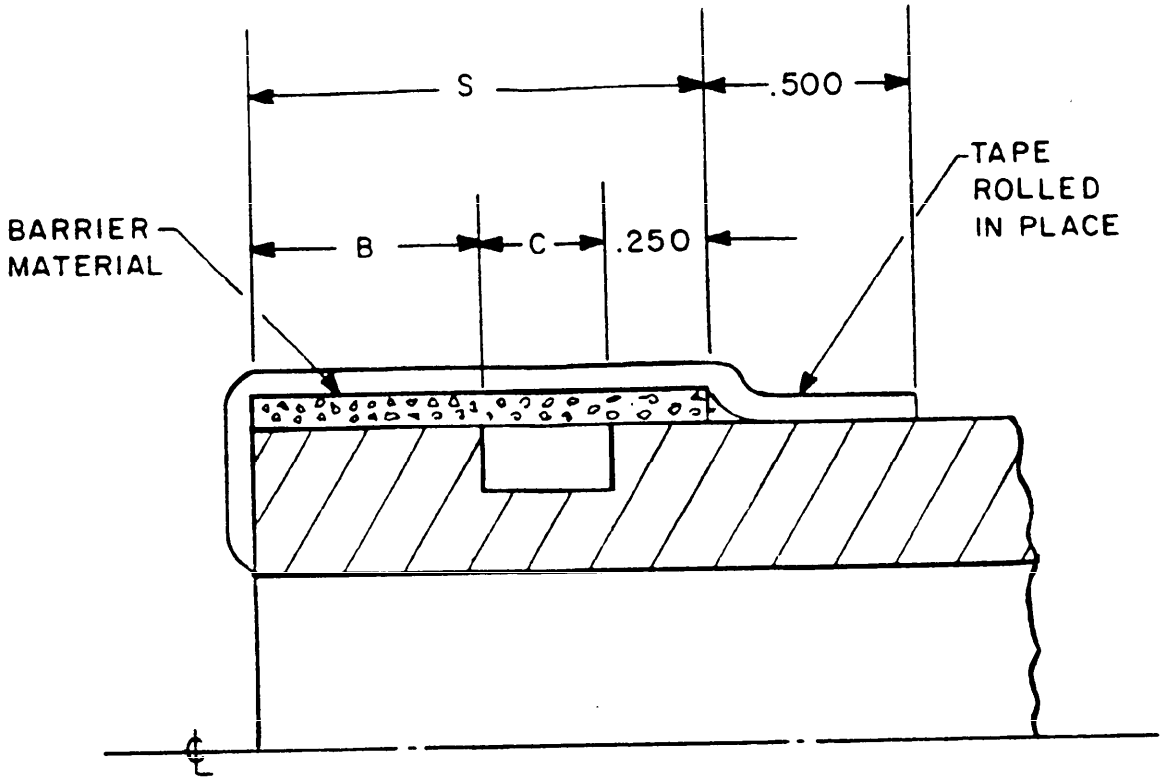
Review activity:
DLA - CS

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 4730-0924

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S ——— LAND SEALING SURFACE

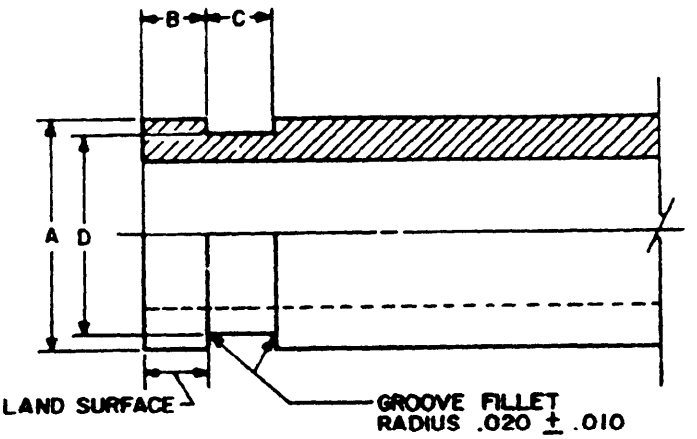
B ——— LAND

C ——— GROOVE

FIGURE 1. Land sealing surface for grooved end pipe, tube end fittings.

X-449B

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NOM PIPE SIZE	GROOVE DIMENSION (INCHES)				
	A		B	C	D
	ALUMINUM	STEEL, MALLEABLE IRON, OR DUCTILE IRON			
3/4	1.065 +0 −.016	1.050 +.031 −.016	0.625 ±.031	0.313 ±.031	0.938 +0.000 −0.015
1	1.330 +0 −.016	1.315 +.031 −.016	0.625 ±.031	0.313 ±.031	1.190 +0.000 −0.015
1-1/4	1.675 +0 −.016	1.660 +.031 −.016	0.625 ±.031	0.313 ±.031	1.535 +0.000 −0.015
1-1/2	1.915 +0 −.016	1.900 +.031 −.016	0.625 ±.031	0.313 ±.031	1.775 +0.000 −0.015
2	2.407 +0 −.016	2.375 +.031 −.016	0.625 ±.031	0.313 ±.031	2.250 +0.000 −0.015
2-1/2	2.905 +0 −.016	2.875 +.031 −.016	0.625 ±.031	0.313 ±.031	2.720 +0.000 −0.015
3	3.532 +0 −.016	3.500 +.031 −.016	0.625 ±.031	0.313 ±.031	3.344 +0.000 −0.018
3-1/2	4.040 +0 −.016	4.000 +.031 −.016	0.625 ±.031	0.313 ±.031	3.834 +0.000 −0.020
4	4.532 +0 −.016	4.500 +.031 −.016	0.625 ±.031	0.375 ±.031	4.334 +0.000 −0.020
5	5.620 +0 −.016	5.563 +.031 −.016	0.625 ±.031	0.375 ±.031	5.395 +0.000 −0.022
6	6.657 +0 −.016	6.625 +.031 −.016	0.625 ±.031	0.375 ±.031	6.455 +0.000 −0.022
8	8.657 +0 −.016	8.625 +.031 −.016	0.750 ±.031	0.437 ±.031	8.441 +0.000 −0.025
10	10.850 +0 −.032	10.750 +.047 −.031	0.750 ±.031	0.500 ±.031	10.562 +0.000 −0.027
12	12.797 +0 −.032	12.750 +.047 −.031	0.750 ±.031	0.500 ±.031	12.531 +0.000 −0.030
14	14.870 +0 −.032	14.000 +.047 −.031	0.938 ±.031	0.500 ±.031	13.781 +0.000 −0.030

NOTE:
ALL DIMENSIONS ARE IN INCHES

FIGURE 2. Dimensions for grooved ends for fittings with grooved backup shoulder.

X-2256A

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